

Sparse Metric Anchors for a Single-View 3D Generative Prior: The Output Frame Is the Bottleneck

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Abstract—A single-view 3D generative model supplies a strong shape prior; a robot supplies a few metric measurements — a handful of contacts, one depth map. We ask what limits the injection of such sparse measurements into a pretrained prior. Of the measurements themselves, only how widely they cover the surface matters much, and a grasp does not choose that; their number, their normals and which face of the object they touch matter little. What limits them once drawn is the *output frame* of the generator. On a billion-parameter wavelet-latent diffusion model we show that the pose of the generated shape is not a function of the input image: it is resampled with the noise seed, shifting by about 135° in median across seeds on 42 YCB objects. Two anchoring routes are then measured against this fact. Injecting contacts as a differentiable constraint during sampling, with no training, improves F-Score@2% by +3.2 on 42 YCB objects (two independent campaigns: +3.17, $p = 9.5 \times 10^{-3}$, and +3.21, $p = 3.1 \times 10^{-4}$; sign test), against +7.9 when the frame is given: frame estimation consumes 60% of the achievable gain, and, once the frame is frozen, the anchored pass is as reproducible as the unanchored one on every object tested. Anchoring the frame instead by reprojecting one depth map into the canonical views of a multi-view variant fixes the pose to $\pm 2^\circ$ and gains +11.2 on average — but the effect is bimodal, losing -22 on solids of revolution, and no ground-truth-free selector we tested recovers the 72.2 that a per-object oracle reaches. Stacking contacts on the anchored frame adds +0.38. Across seven controlled manipulations and a 2×2 design over 126 paired cases, what an anchor set needs is spatial spread and placement accuracy: at equal count, uniform coverage gains twice what eight contact patches gain, while count, surface normals and which face they lie on matter little — 128 depth-like points on the visible face gain +15.6 against +7.9 for eight tactile patches on the hidden face. Five alternative fusion formulations that treat the measurements as a generator all fail from that single cause, which we quantify without any diffusion model: six oriented contacts reconstructed by Poisson reach 19.9 F@2 % against 56.5 for the image alone.

I. INTRODUCTION

Large pretrained 3D generative models reconstruct a plausible object from a single image. A robot holding that object knows things the image does not: where its fingers are, and the depth of what the camera sees. The natural question is how to inject such measurements into a model that was never trained to receive them.

This paper started as a study of touch — whether a handful of tactile contacts can improve a single-image reconstruction — and its central finding is that the question was framed on the wrong axis. Contacts do help, modestly and significantly. But what decides how much they help has little to do with what a hand can vary: not their number over a $16\times$ range,

not whether they carry a surface normal, and only marginally whether they lie on the face the camera cannot see. Two things decide. How widely the anchors cover the surface, which a grasp does not control; and whether the measurements can be placed in the frame in which the generator produces its shape — and that frame, we show, is not a function of the input. It is drawn with the noise.

We therefore treat the problem as one of *anchoring* a prior with sparse metric measurements, and use two sources of measurement as two probes of the same bottleneck. Tactile contacts enter as a differentiable constraint applied during sampling, with no training. A single depth map enters as a conditioning signal, reprojected into the canonical views of a multi-view variant of the same backbone. The first cannot fix the frame and pays for it; the second fixes the frame but does so by handing the model evidence it treats as real, with object-dependent consequences. Neither is a solution. Together they locate the problem.

A. Contributions

- 1) The **output pose of a single-view 3D diffusion model is not learnable from its input**: across three noise seeds on the same image, it moves by about 135° in median on 42 objects, with asymmetric objects no more stable than symmetric ones (Sec. III).
- 2) A **training-free constraint-anchoring scheme**, differentiable end-to-end at sampling time, and a measurement of what the frame costs it: +3.17 realised against +7.91 with the frame given, on shared unanchored passes, replicated by an independent campaign at +3.21 / +7.82 (Secs. IV-A, VI-A).
- 3) A **conditioning-anchoring scheme** by depth reprojection, which fixes the frame and gains +11.2, together with the finding that the gain is bimodal and that the obvious ground-truth-free selector fails (Secs. IV-B, VI-B).
- 4) A **characterisation of what an anchor set must be**: seven controlled manipulations and a 2×2 design over 126 paired cases show that spread and placement accuracy matter — spread alone doubles the gain — while count, normals and which face the anchors lie on matter little; the depth-camera control, points covering the visible face, outgains tactile patches on the hidden face (Sec. VI-C).

- 5) A **decomposition of run-to-run variance**, showing that with the frame frozen the guidance adds no variance on any of four objects, so what dominates is frame re-estimation, and three evaluation pitfalls that each silently reversed a conclusion (Sec. VI-F).

II. RELATED WORK

Visuo-tactile shape reconstruction. Prior work combines the two modalities either by fusing learned representations of both [7], [8] or by accumulating dense contacts into an implicit surface [9]. Both assume enough tactile data to describe a shape: [7] learns a chart-based fusion from simulated touch across a large array of objects, and [9] integrates dense Gel-Sight readings into a Gaussian-process implicit surface. Our regime is the opposite. A grasp from a nine-zone hand returns single-digit contact counts, too few to fit any surface; their value is as anchors on a prior, not as a signal to reconstruct from. Sec. VI-D shows that formulations which treat them as a generator fail for one shared reason.

3D generative models. We build on WaLa [1], a billion-parameter wavelet-latent diffusion model that predicts $\hat{z}_0$ directly and decodes it to a 256^3 signed distance field. Its single-view variant is our main backbone; its four-view depth variant is the vehicle of Sec. IV-B. We retrain nothing.

Guidance in diffusion models. Classifier guidance [2] and classifier-free guidance [3] steer sampling toward a condition. Our constraint scheme is closest to the former in spirit but uses no classifier: the correction is the analytic gradient of a geometric residual, so it has no learned parameters and no training distribution to leave.

Pose-conditioned generation. Zero-1-to-3 [4] conditions on an explicit relative camera transform; feed-forward stereo methods such as DUST3R [5] regress pose directly from image pairs. Editing-style initialisation such as SDEdit [12] offers a third route, which we discuss but do not test. Sec. III explains why none is a drop-in fix here: the single-view backbone receives no camera information at all, and the variants that do receive it destroy the very margin the measurements were meant to fill.

III. THE OUTPUT FRAME IS NOT A FUNCTION OF THE INPUT

Measurements live in the sensor frame; the generator produces in a canonical frame of its own. Any anchoring scheme must map one onto the other. Before asking how well, we ask whether the target of that mapping exists.

The cause, in one configuration line. The single-view backbone declares `use_camera_index = False`: the function that extracts a view index returns `None`, and the model receives no information about the camera at all. There is therefore no frame convention to recover; the output pose is whatever the content of the image and the noise produce.

The pose is resampled with the noise. The cheapest imaginable fix would be to regress the output rotation from the image, with no retraining of the generator. It presumes that the pose is a function of the image. Comparing the rotations

obtained for three noise seeds on the *same* image, over 42 objects, the median inter-seed discrepancy is $136.5^\circ \pm 3.7$ for asymmetric objects and $135.7^\circ \pm 3.7$ for symmetric ones, mean and standard deviation over four independent runs of the registration; in every run, 1 of 16 asymmetric objects is stable below 30° and 0 of 26 symmetric ones; the asymmetry/stability correlation is -0.14 ± 0.06 . The registration samples points at random, and on a solid of revolution the rotation it finds is arbitrary, which is why the medians carry a spread; the split-level conclusion does not — the stable counts and the absence of any asymmetry–stability relation are identical in all four runs. (The per-object asymmetry measure itself is stable to 0.001; one object sits on the 0.02 threshold and changes side in one run of four, giving 17/25 there.) The split by symmetry rules out a measurement artefact — two identical outputs of a solid of revolution register with very different rotations — and the asymmetric objects, whose orientation is identifiable, are *no more stable*. No regressor can learn this pose: not for want of the right estimator, but because the target does not exist.

Models that receive the pose anchor it. The multi-view variants of the same backbone declare `use_camera_index = True` and take four images with their index in a canonical 55-position rig. Their prediction-to-reference rotation is reproducible to 0.8° across seeds and identical across well-constrained objects. But four views also solve the reconstruction — up to 99.9 F@2% on objects the single-view model scores at 23 to 59 — leaving nothing for a measurement to add.

Pose and margin do not coexist in any published variant. This is the tension the two anchoring routes below have to negotiate.

IV. TWO WAYS TO ANCHOR

A. Anchoring by constraint during sampling

The backbone predicts $\hat{z}_0$ directly rather than the noise residual, so the network output *is* the current shape estimate and can be corrected without touching the noise schedule or the sampler. At each denoising step we decode $\hat{z}_0$ to a dense signed distance field $\Phi(\hat{z}_0) \in \mathbb{R}^{256^3}$, evaluate it at the N anchor points p_i , and take a step along the gradient of

$$\mathcal{L}_{\text{pos}} = \frac{1}{N} \sum_{i=1}^N \left[\mathcal{T}(\Phi(\hat{z}_0), p_i) \right]^2, \quad (1)$$

where $\mathcal{T}$ is trilinear interpolation. The imposed constraint is simply *the surface passes through these points* (Fig. 1). The step is normalised, $\Delta = w \|\hat{z}_0\| \nabla \mathcal{L} / \|\nabla \mathcal{L}\|$, so that the weight w depends on neither the scale of the distance field nor that of the latent, both of which drift during sampling.

The guidance never needs a mesh, only the field: marching cubes [11] is applied once at the end, to export the result. Crucially, *no prediction is ever blended with another*. The destructive interference that condemns the composition-based formulations of Sec. VI-D cannot occur.

Implementation. $\mathcal{T}$ is written by hand as indexing and products rather than with `grid_sample`: the 3D backward

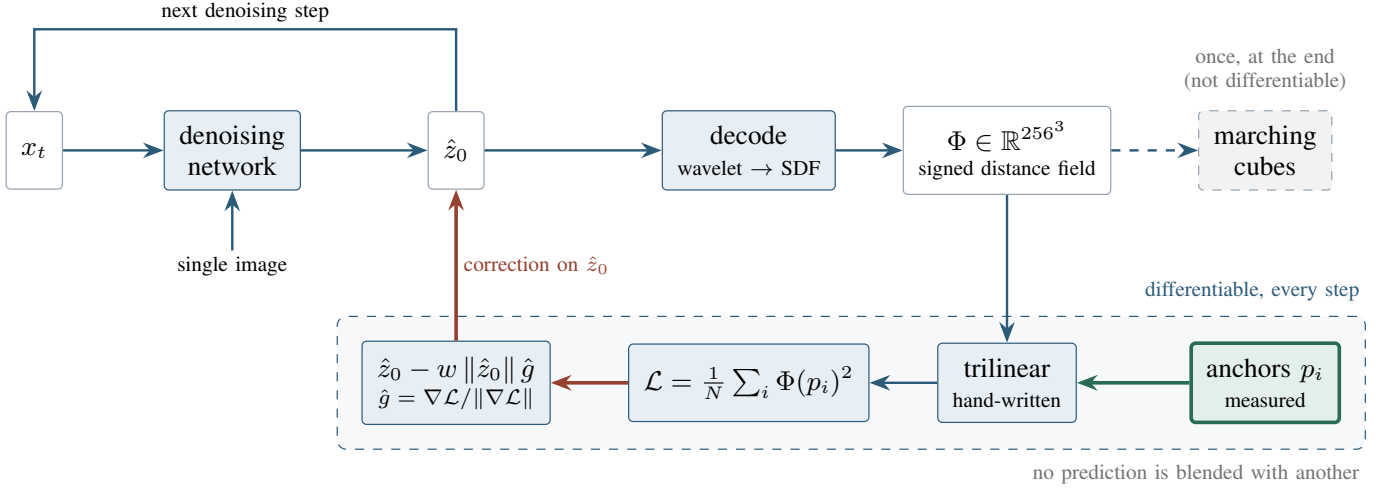


Fig. 1. **Constraint anchoring, in place.** The correction acts on the predicted $\hat{z}_0$ *inside* the denoising loop, through a decode that is differentiable end to end; the anchors are the only measured quantity in the diagram (green). Marching cubes sits outside the loop because it is not differentiable — and the guidance never needs a mesh, only the field. Because no prediction is ever blended with another, the destructive interference that condemns the composition-based formulations of Sec. VI-D cannot occur.

kernel of the latter is unimplemented on Apple Metal, and the CPU fallback would impose a 67 MB round trip per step. Measured cost: 3.44 s per guided step, 8.98 GB peak memory, ≈ 55 s per sample at 20 steps on an M2 Max — no remote compute. The backbone’s weights are used unchanged, but its code is not the published one. Running it on Metal required casting float64 tensors to float32 at several points and building the sparse-convolution library natively; a backward pass through the model fits in memory only with gradient checkpointing on the diffusion transformer (peak 21 $\rightarrow$ 6.5 GB); and three pull requests of the public repository are merged in. The guidance itself wraps the denoiser at sampling time and changes neither the sampler nor the weights. The anchoring campaigns reported here all ran on that port, and none was cross-checked against the reference CUDA implementation. The modified code is released with the paper, with a notice listing these changes as the backbone’s licence requires.

Optional surface-normal term. A contact from a hand carries an approximate outward normal n_i . We impose only the *direction* of the field gradient, $\mathcal{L}_{\text{nm}} = \frac{1}{N} \sum_i [1 - s g_i n_i / \|g_i\|]$ with $g_i = \nabla_p \Phi|_{p_i}$ by central finite differences and $s = +1$ the measured sign convention (exterior positive; interior median -0.0998 , exterior $+0.0999$, a TSDF truncated at ± 0.10). The two losses have incommensurable units; λ is calibrated once at the first guided step so that their gradient norms match, then frozen. Sec. VI-C shows the term does not measurably help.

Placing the anchors. Without ground truth, the frame is estimated from two criteria: *silhouette consistency* (project the prediction and compare to the conditioning image, which the model already received) and *anchor residual* (among silhouette-plausible poses, keep the one whose anchors lie closest to the first-pass surface). Neither suffices alone. Registering anchors onto an unguided first pass by ICP places them, against their exact position, at 0.66–0.76 in median in a

frame where the object spans $[-1, 1]$ — *on* the surface but in the *wrong place*, since ICP superimposes without respecting correspondences and near-symmetries leave it that latitude. Silhouette plus residual brings the median gap from 0.755 to 0.501, and to 0.111 on objects without silhouette ambiguity.

B. Anchoring by conditioning through a multi-view variant

The four-view depth variant anchors the frame but demands four inputs, and four real views leave no margin. Feeding one real view and three blank images fails: measured on four objects and two seeds, 43.07 F@2 % against 95.40 with four real views, and worse than the dedicated single-view model on half the objects. **Blank views are not neutral; they remove information.**

A depth map, however, is 3D geometry in the camera frame. It can be back-projected to points and re-rendered from any other position of the canonical rig. The synthetic view is geometrically correct where it has data and empty elsewhere — far closer to the training distribution than a black image. We therefore feed the model *one real depth view and three reprojected from it*, with their four camera indices. The placement of anchors then becomes a fixed transform, calibrated once: measured on four objects and two seeds, the output rotation stays between 118° and 122° about a constant axis.

The design intent was that the reprojection *add no information* — the same single observation re-encoded — so that reconstruction would stay at the single-view level while the camera indices anchor the pose. Sec. VI-B shows that this intent was not met: the model treats the reprojected views as evidence, and what it does with their holes depends on the object.

C. Anchor model

Sampling anchors uniformly on the reference mesh is the heaviest objection against any anchoring result, since the points then come from the answer. We replace it with a palpation model imposing three physical constraints: straight-line reachability (rays cast from outside, first hits only), restriction to the face hidden from the camera that rendered the image (85% of the anchors are hidden from it, 25 to 98% depending on the object; an earlier draw from a misplaced camera reached 61%, see Limitations), and rejection of points where a 6 mm fingertip would not fit. Anchors are grouped into *sites* rather than scattered — a hand touches in patches. This change alone halves the ceiling of the method ($+14.37 \rightarrow +6.77$ on five objects): part of the gain reported by intermediate campaigns came from uniform coverage, not from the measurements. For Sec. VI-C the same model is also run on the visible face, and uniform draws are taken on either face, so that spread and face can be crossed.

For Sec. VI-E we additionally use a *grasp simulator* that places one anchor per finger zone (thumb, index, middle, ring, little, palm) at fixed anatomical angles around the object’s principal axis, far closer to a real hand than well-spread sites: minimum inter-anchor distance 0.058–0.344 against 0.446–1.281 for our sites.

V. EXPERIMENTAL SETUP

Protocol. 42 YCB objects [6], 3 noise seeds each, giving 126 *paired* comparisons: each anchored generation is compared to an unanchored one with the *same seed*. Because nothing is trained, every object is a test object.

Metric. F-Score@2% after alignment to the reference, plus symmetric Chamfer distance and connectivity. For the anchor-set design of Sec. VI-C, also the recall of the reference surface hidden from the rendering camera and of the surface it sees, split by hidden-point removal.

Significance. We use a **sign test**, not a *t*-test: the gain distribution is strongly skewed (median +1.69 for a mean of +3.21) and a few objects dominate. A mean is fragile there; a count of signs is not. All *p*-values are **two-sided**, and exact ties are discarded rather than counted as failures — one case out of 126 scores identically with and without anchoring, which is why the main result is reported on 125.

Oracle discipline. Where stated, anchors are placed using the reference mesh, drawn with the palpation model of Sec. IV-C. Such numbers are reported *only* as upper bounds, never as system performance, and always with an explicit statement of what was given for free.

VI. RESULTS

A. Anchoring by constraint: what the frame costs

Table I gives the constraint result. The effect is modest relative to its spread — it is the significance of the *sign* that is solid, not the precision of the magnitude. On five objects an earlier campaign measured +1.79 at $p = 0.035$; at 42 objects the effect is stronger, so it was not a single object carrying everything. It also replicates: an independent

TABLE I
CONSTRAINT ANCHORING WITH TACTILE CONTACTS, 42 YCB OBJECTS $\times$ 3 SEEDS. BOTH ARMS ANCHOR THE SAME UNANCHORED PASSES. LAST ROW: AN EARLIER, INDEPENDENT CAMPAIGN WHOSE PALPATION SITES WERE DRAWN FROM A MISPLACED CAMERA (LIMITATIONS).

Unanchored baseline (image only)	52.7 F@2 %
Mean gain, frame estimated	+3.17
Median / std. dev.	+1.62 / 8.18
Positive cases	78 / 126
Sign test (two-sided)	$p = 9.5 \times 10^{-3}$
Objects with positive mean gain	30 / 42
Mean gain, frame given (oracle)	+7.91
Positive cases	111 / 125
Frame cost, paired (oracle — estimated)	$+4.74 \pm 0.68$
Replication, earlier draw: estimated / oracle	+3.21 / +7.82

TABLE II
THE TWO ANCHORING ROUTES, AND THEIR COMBINATION. SAME 42 OBJECTS, SAME 126 SEED-PAIRED CASES, SAME METRIC.

	F@2 %	frame
Image only (single-view backbone)	53.1	drawn with noise
+ contacts as constraint	56.3	estimated, $\pm$ large
Depth reprojected to 4 views	64.3	fixed, $\pm 2^\circ$
+ contacts as constraint	64.7	fixed
Per-case oracle choice of route	72.2	—
Four <i>real</i> depth views (4 objects)	98.6	fixed

42-object campaign, with its own unanchored passes and palpation sites drawn from a misplaced camera (Limitations), measured +3.21 (83/125, $p = 3.1 \times 10^{-4}$) and +7.82 with the frame given; case by case, the two campaigns differ by -0.04 ± 0.70 and $+0.10 \pm 0.63$.

Comparing the realised gain to the oracle isolates the cost of estimating the frame. Both arms anchor the same unanchored pass, so the comparison is paired on the anchored scores: $+4.74 \pm 0.68$ (102/126, $p = 10^{-12}$), i.e. the estimated frame realises 40% of the ceiling; under the five-object sweep campaign, +1.79 against +6.77. **Frame estimation absorbs about 60% of what constraint anchoring can deliver from a hand’s contacts** — more than any other factor a hand controls (Sec. VI-C).

B. Anchoring by conditioning: the frame fixed, at a price

Table II places the two routes side by side. Reprojecting a single depth map into the four canonical views gains +11.22 over the single-view backbone in paired mean — more than three times what contacts deliver — and fixes the frame. But the gain is **bimodal**: 87 of 126 cases improve by +26.1 on average, and 39 cases lose -22.1 . The losses are catastrophic on solids of revolution — baseball $95.5 \rightarrow 29.2$, soccer ball $93.9 \rightarrow 35.5$, master chef can $91.0 \rightarrow 46.8$ — where a single depth view leaves three canonical views almost empty and the model fills them from its prior, badly. The design intent of Sec. IV-B, that reprojection add no information, was not met: the model treats the reprojected views as evidence.

TABLE III

ANCHOR SET, 2×2 : GAIN IN F@2 % OVER ONE SHARED UNANCHORED PASS, ORACLE FRAME, 42 OBJECTS $\times$ 3 SEEDS, 128 ANCHORS IN EVERY CELL. IN BRACKETS, THE FRACTION OF EACH SET HIDDEN FROM THE RENDERING CAMERA.

	hidden face	visible face
8 palpation patches	+7.91 [0.85]	+7.31 [0.06]
Uniform coverage	+18.24 [0.64]	+15.57 [0.01]
Spread: uniform – patches	+10.33 (115/125)	+8.26 (104/124)
Face: hidden – visible, patches	+0.60, $p = 0.47$	
Face: hidden – visible, uniform	+2.67, $p = 9 \times 10^{-7}$	

No ground-truth-free selector we tested recovers the oracle. A per-case choice between the two routes would reach 72.2. The obvious selector — the fraction of reprojection pixels that carry depth — is *anti*-correlated with the gain ($r = -0.40$): compact round objects fill many pixels and fail, thin flat ones fill few and succeed. Coverage measures object shape, not reprojection quality, and no threshold on it beats always choosing reprojection. Symmetry classification, the other candidate, was already refuted as a predictor in Sec. VI-C. The 72.2 exists but is out of reach without knowing the answer.

Stacking contacts on the anchored frame adds +0.38 (75/126 positive, $p = 0.040$). Once the frame is fixed and the prior is strong, contacts have little left to correct. The two routes are not complementary either: the correlation between the reprojection gain and the contact gain is +0.056, and on the 39 cases where reprojection fails, contacts recover +1.7 against a loss of −22. They fail on different objects by accident, not by design.

C. What an anchor set needs

Fig. 2a sweeps seven properties of the anchor set under constraint anchoring, on five objects (42 for the visible-face row), oracle placement. Three clear a two-sided sign test, and they share a theme.

Spread matters. At constant budget of 128 anchors distributed over N sites, the gain is −0.42 at 4 sites, +1.79 at 8, +2.03 at 16 and +1.55 at 32: a threshold between 4 and 8, then a plateau. Going from 8 patches to 4 costs −2.55 (12/15 negative).

Placement accuracy matters. Gaussian noise of 3mm on position and 15° on normals, the order of what a finger’s forward kinematics would deliver, costs −2.40 (12/15 negative).

Count does not. From 32 to 512 anchors, the gain is +0.23, +1.21, +0.07: flat over a $16\times$ range.

Normals do not. Adding them gives +0.93, which the sign test cannot separate from zero (10/15, $p = 0.302$). The mean is carried by two outliers: dropping the largest leaves +0.43, against a median of +0.26.

Spread dominates, and a hand does not choose it. A 2×2 design over the 42 objects and three seeds, oracle frame, one shared unanchored pass (Table III), crosses how the 128 anchors spread — eight palpation patches, or uniform coverage — with the face they lie on. At equal count, uniform coverage

TABLE IV

SIX ORIENTED CONTACTS, WITH AND WITHOUT A GENERATIVE PRIOR. POISSON VALUES ARE MEAN $\pm$ S.D. OVER FIVE RUNS.

Poisson reconstruction [10], 6 oriented contacts	19.90 $\pm$ 0.36 F@2 %
Poisson reconstruction, 18 contacts (3 grasps)	20.44 $\pm$ 0.18
Generative prior, image only, no measurement	56.52

gains +10.3 more than patches on the hidden face and +8.3 more on the visible face ($p < 10^{-14}$ both): spread alone doubles the gain, and a grasp is patches. The face matters little in comparison: nothing between patches (+0.6 $\pm$ 0.6, $p = 0.47$), +2.7 $\pm$ 0.5 in favour of the hidden face under uniform coverage ($p = 9 \times 10^{-7}$; that draw is only 64% hidden, so this is a lower bound). Where the gain lands follows the same pattern. With visibility decided from the rendering camera, hidden-face anchors improve the recall of the hidden side more than visible-face anchors do — +6.7 $\pm$ 1.0 under uniform coverage, +1.2 $\pm$ 0.8 with patches — and the visible side about equally. The premise thus survives at the level of recall, by a margin the spread effect dwarfs. The grasp simulator agrees: three strategies placing the same six anchors with 87%, 60% and 30% of them hidden gain +1.37, +1.73 and +2.55, none separable from the others, and over 45 cases the correlation between hidden fraction and gain is $r = -0.001$ (Fig. 2b).

The depth-camera control. The visible-face, uniform cell is what a depth camera delivers: 128 points on the face the camera sees, none on the face it cannot. It gains +15.6 — twice the +7.9 of eight tactile patches on the hidden face (+7.7 $\pm$ 1.0 paired, 103/125, $p = 10^{-13}$). The constraint route therefore needs no touch to work; touch would outgain depth only where it covered the object as widely, which one grasp does not. Sparse anchors do not complete what vision misses. **They fix global position, scale and orientation of a prior whose frame is loose, and for that, covering the surface matters far more than which side they lie on.**

D. Why fusion formulations fail

Five formulations require the measurements to produce a complete shape that is then blended with the image’s. The most instructive failure is weighted blending in $\hat{z}_0$ space, which collapses from 44.9 to 13.9 F@2 % at equal weight: convex interpolation between two distinct modes of a distribution lands in a low-density region — the midpoint between two plausible shapes is not a plausible shape. This holds for any naive blending of generative latents.

Table IV converts the shared cause into a measurement without any diffusion model. Poisson reconstruction consumes exactly oriented points and yields a watertight surface in 9 of 10 cases — this is not gravel. It still reaches only 35% of what the image alone already achieves. Sparse measurements are a poor generator and a good anchor; the division of labour is not a design choice but what the data imposes.

A *learned* fusion module (a geometry encoder and cross-attention grafted onto the frozen backbone, 50 epochs, 5–10M

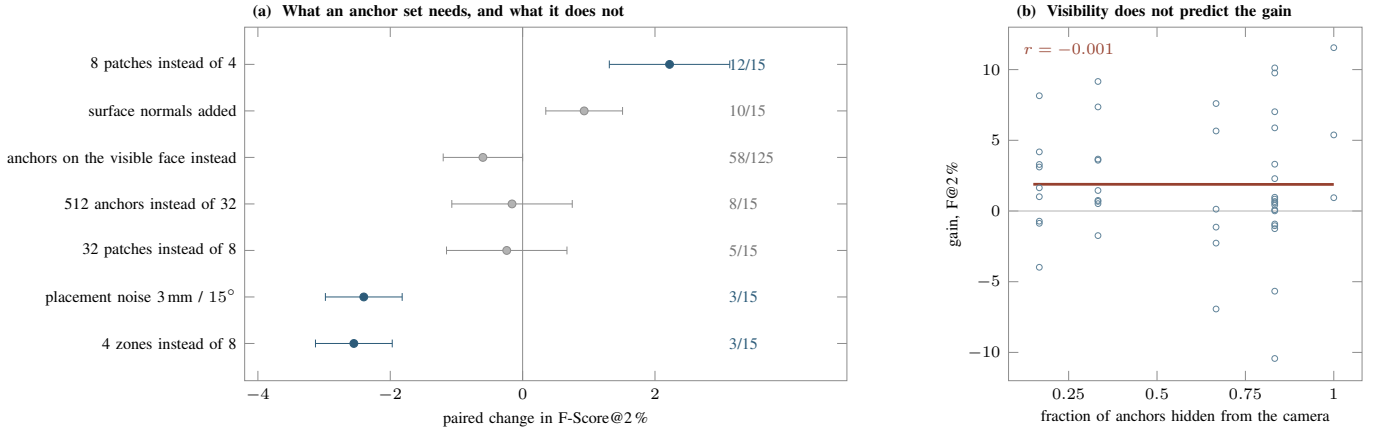


Fig. 2. **Seven controlled manipulations of the anchor set.** (a) Every estimate is a paired difference between two anchored conditions on the same object and the same noise seed ($n = 15$; the visible-face row comes from the 2×2 design of Table III, $n = 126$, with its own standard error of 0.60). Error bars are one standard error — 0.58 when both conditions come from the same campaign and therefore share the oracle frame, so that placement noise cancels in the difference (pooled over four objects), and at least 0.91 when they come from different campaigns, which share no unanchored pass (measured on one object). Colour marks significance under a two-sided sign test, whose counts are given at the right; we rely on it rather than on the standard errors, because it needs no estimate of σ . Only three comparisons clear it, and all three concern *where* the anchors are and *how accurately* they are placed — never how many there are, nor what the camera can see of them. (b) The grasp-simulator result on its own, over the 45 cases of the grasp-strategy campaign, with visibility decided from the camera that rendered the images. The least-squares fit is flat.

TABLE V
HARDWARE-REALISTIC CARDINALITY, ORACLE PLACEMENT. PAIRED DIFFERENCES BELOW.

Condition	Mean gain	Sign test
8 anchors, position only	+3.25	$p = 0.118$
8 anchors + normals	+4.18	$p = 0.118$
8 anchors, noisy placement	+1.78	$p = 0.302$
4 anchors only	+0.70	$p = 1.000$
<i>Paired differences (same object, same seed)</i>		
Normals vs. none	+0.93	$p = 0.302$
Placement noise 3 mm / 15°	-2.40	$p = 0.035$
4 zones instead of 8	-2.55	$p = 0.035$

trainable parameters) measures -4.20 over five objects with unpaired draws — 0.9 standard errors, indistinguishable from zero. The comparison with constraint anchoring is not really about the two scores, both poor. It is that the learned route **consumes the objects it would need in order to judge itself**: 15 of 20 become training data, leaving five to conclude on, with unpaired noise four times larger. In a small-data, noisy-metric regime a method that needs no training buys *statistical power*, and that decided this study.

E. What survives at hardware cardinality

A nine-zone dexterous hand excites only three to five zones in a real grasp. At one anchor per zone (Table V), a $16\times$ reduction from the reference campaign costs roughly half the ceiling ($+6.77 \rightarrow +3.25$), not all of it. But the two effects that reach significance are precisely the two properties of a real grasp: placement noise and reduced zone count. A real hand has both, and the joint condition — clustered simulator anchors, with normals, placement noise *and* estimated rather

than oracle frame — has not been measured. We record the prediction, before measurement, that it will be null.

F. Methodological results

Run-to-run variance, and where it comes from. Re-running an identical command eight times — same object, same seed, same anchors — gives a standard deviation of 0.49 for the unanchored pass but 2.29 for the anchored one, and 2.50 for the paired difference. The obvious reading, that guidance amplifies GPU non-determinism, is *wrong*, and we report it because we held it first. Those repetitions are not identical: the oracle frame is re-estimated from each repetition’s own unanchored pass, so the anchor positions move. Freezing the frame separates the causes:

	re-estimated frame	frozen frame
unanchored	0.49	0.68
anchored	2.29	0.70
paired difference	2.50	1.29

With the placement frozen, the anchored pass is as reproducible as the unanchored one (0.70 against 0.68): on this object, **91% of the variance of an anchored score comes from re-estimating where the anchors go**. Repeating the frozen-frame measurement on three more objects gives anchored standard deviations of 1.74, 0.40 and 2.56 against unanchored ones of 2.53, 0.49 and 2.19 — the anchored pass is never noisier than the unanchored one, on any object. **Guidance amplifies nothing.** What does vary is the object: the floor spans 0.40 to 2.56 across the four, a factor of six. Much of it is the scorer, not the generator. Re-scoring each saved mesh four times, without generating anything again, gives standard deviations of 0.33 to 2.23 depending on the object. Pooled over the four, the scorer alone has a standard deviation

of 1.45 where an anchored score has 1.60, and 1.03 where an unanchored one has 1.72: both the alignment and the F-Score sample points at random, and the ICP occasionally settles in another basin, up to 8 points apart on the same mesh. With the frame frozen, generation itself adds little (0.66 at most, by difference). Our standard errors come from single scorings and therefore include this noise; averaging several scorings per prediction would shrink them. The frame problem of Sec. III therefore appears twice — it consumes most of the mean gain, and it dominates the variance.

This fixes the error model. Within one campaign the oracle frame is computed once per (object, seed) and shared by every condition, so placement noise cancels in condition-versus-condition differences; across campaigns it does not, since no two share an unanchored pass. Pooled over the four objects, the frozen-frame σ of an anchored score is 1.60, which puts the within-campaign standard error of a paired difference at 0.58 over 15 cases; the cross-campaign figure, 0.91, was measured on one object only and is a lower bound (Fig. 2a). Under these, the two degradations sit at 4 standard errors and more, and the two effects the sign test rejects — normals, visibility — at 1.4 to 1.6. We nonetheless base every claim on the sign test, which needs no estimate of σ and gives the same verdict.

F-Score is blind to fragmentation. A scatter of disjoint shards near the true surface scores well. On the banana: a 128-point cloud input yields 43 connected components with 11% of faces in the largest, and 47.83 F@2 %, against 34.48 for an image-only prediction that is a single closed surface. We report the number of components and reject distance metrics below a threshold on the largest-component ratio.

F-Score is blind to mirror symmetry. Building registration candidates from SVD bases yields orthogonal but not necessarily right-handed frames: **5 of 15 registrations were reflections**. Fixed by forcing the determinant to +1.

A badly initialised ICP fails silently. Identity-initialised ICP leaves the registration outside its basin of convergence, understating one object by up to 31 points with no error signal. This changed the ranking between compared methods, and was the source of the learned module’s original “−8 points” verdict.

VII. LIMITATIONS

A camera placed in the wrong frame. Blender’s OBJ importer turns each Y-up mesh into its Z-up world, $(x, y, z) \mapsto (x, -z, y)$, before the camera at $(2.2, -2.2, 1.8)$ renders it. Our first occlusion computations placed that camera in the mesh’s own frame instead, 68° from where it rendered: projected from there, the reference silhouettes overlap the input images at 0.51 on average over 42 objects, against 0.95 with the turn applied, and the visible/hidden label differs on 26% of the surface. We found it late, and redrew. Table I, Table III and the recall of Sec. VI-C use anchors drawn from the rendering camera; the grasp-strategy fractions of Fig. 2b are recomputed from it (from the misplaced camera the correlation read −0.145, and three cases seemed to leave no anchor on the hidden face). Two artefacts keep the error. The five-object sweeps of Fig. 2a

and Table V use palpation sites that are 61% hidden from the rendering camera rather than 85%; every comparison there is between two sets drawn the same way, so the error sits on both sides of each difference. And the 42-object campaign reported as a replication in Table I used those same sites; that its gains match the redrawn ones to -0.04 ± 0.70 says that a mostly hidden set and a hidden-face set behave alike, which Table III confirms directly. A smaller defect adds to it: the renderer normalises objects to a maximum dimension of 1.3 where the occlusion code uses 2.0; with the camera corrected, it changes the label of 3 of the 45 grasp cases.

Two caveats of the 2×2 design. Its uniform hidden-face draw is only 64% hidden, so the face effect under uniform coverage is a lower bound. And on 17 of the 126 cases a few anchors fall outside the generator’s grid after placement and are ignored by the guidance; dropping those cases changes no conclusion (spread +10.5 / +8.4, face +0.7 / +2.7).

Images are synthetic Blender renders under a single camera and lighting condition. Anchors are sampled on the reference mesh — filtered for reachability, occlusion and fingertip clearance, but noiseless and never false, except where noise is added deliberately. No real object, sensor or robot is involved, although the robotic bench that motivated this work exists and is operational. A single backbone family and a single guidance weight were used; three seeds per object is few; secondary experiments use five objects and one of them, the power drill, carries a large share of several totals. Formulations 3, 4 and 7 of the fusion taxonomy are *moot* in this data regime rather than invalidated — early-fusion retraining, the only route the backbone’s authors validate, was out of budget.

VIII. CONCLUSION

A single-view 3D generative prior can be anchored by sparse metric measurements. Of the measurements themselves, one property matters: how widely they cover the surface — uniform coverage gains twice what eight contact patches gain — and a hand does not choose it. Their number does not matter over a $16\times$ range; their normals do not measurably help; which face they lie on matters little, and a depth camera’s points on the visible face outgain tactile patches on the hidden one. What limits a hand’s measurements, once drawn, is whether they can be placed in the frame the generator produces in — and that frame is drawn with the noise, so no estimator can learn it. Estimating it anyway costs 60% of the achievable gain, and it is what varies between runs: with the frame frozen, anchoring adds no variance on any of four objects.

Fixing the frame by conditioning is possible, and worth three times more than constraint anchoring on average, but it is not free: the model treats the reprojected evidence as real, and on a third of the objects the price is catastrophic, with no selector in sight that does not already know the answer. Stacking constraints on the fixed frame adds almost nothing.

The mechanism consistent with all of this is not that measurements complete what vision misses. It is that *a few true anchors regularise the output of a generative prior* — position, scale, orientation — and that the prior’s own frame

is what stands in the way. That is a more modest claim than the one this work set out to make, and a more general one: it applies to touch, to depth — which, covering the surface, does it better — and to any sparse metric signal a robot can bring.

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